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HW Set 04: N-Body Simulations

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ABSTRACT

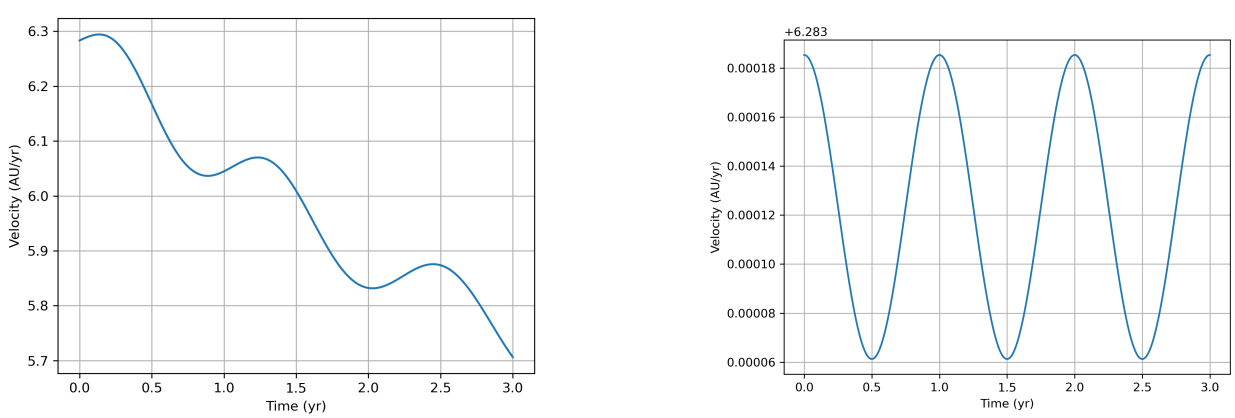
The fourth homework: it's n-body simulation time. All code can be found in the Jupyter Notebook: https://github.com/kparrine/Computational_Astro/blob/main/C4/C4.ipynb.

1. QUESTION 1

For part (a), the animations can be found at these links:

1. Euler: https://github.com/kparrine/Computational_Astro/blob/main/C4/euler.gif
2. Leapfrog: https://github.com/kparrine/Computational_Astro/tree/main/C4/leapfrog.gif

With Earth at a distance $r_0 = 1$ AU and with a velocity given by circular motion, Figure 1 shows the speed of Earth as a function of time for Euler (a) and Leapfrog (b). Technically, the speed for both methods is not constant, since for Euler, the speed slowly decreases over time, while for Leapfrog, the speed fluctuates in a sinusoidal type motion. Note that these fluctuations are very small. However, in the Leapfrog method, this sinusoidal motion cancels out, so the speed is constant. In the comparison plot between Euler and Leapfrog of Figure 2, it looks like Leapfrog has a constant velocity over time due to this "canceling out".



(a) (b)

Figure 1. Speed of Earth as a function of time for three orbital periods integrated with the (a) Euler method and with the (b) Leapfrog method.

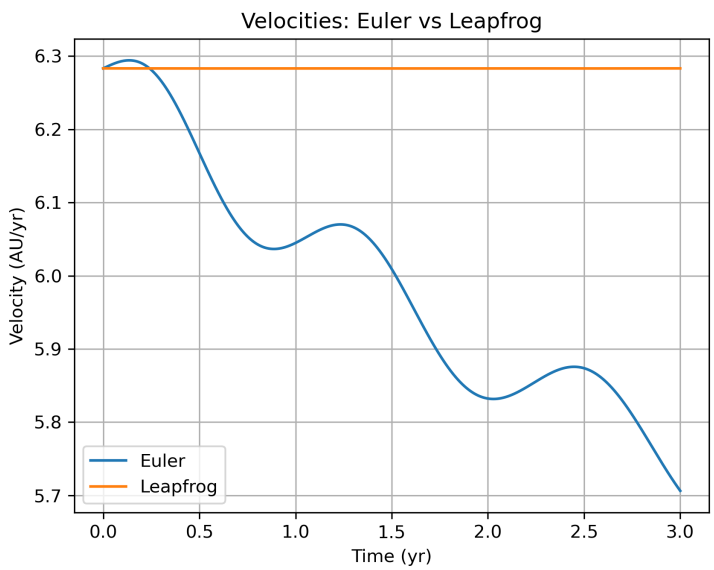
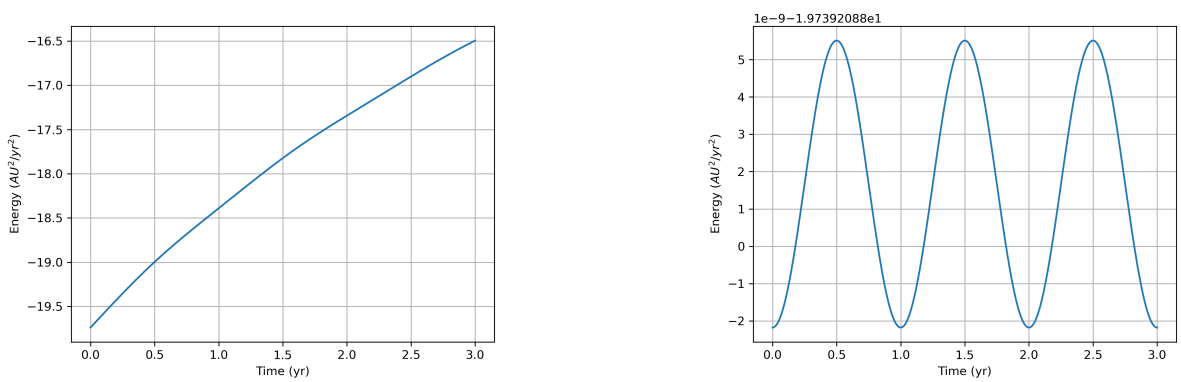


Figure 2. Comparing the speed of Earth as a function of time for three orbital periods integrated by the Euler method (blue line) and by the Leapfrog method (orange line).



(a) (b)

Figure 3. The energy as a function of time for three orbital periods integrated with the (a) Euler method and with the (b) Leapfrog method.

The initial total specific energy for Euler is $-19.74AU^2/yr^2$ and the final total specific energy is $-16.49AU^2/yr^2$. For Leapfrog, the initial and final specific energy is $-19.74AU^2/yr^2$. Figure 3 shows the energy as a function of time for Euler (a) and Leapfrog (b). For Euler, the energy of the orbit increases, so the Earth begins drifting outwards, therefore, the orbit becomes more unstable. For Leapfrog, the behavior is similar to the velocity, where is acts in a sinusoidal motion, therefore, canceling out the errors and no drift is observed. This makes is seem like energy is conserved in the comparison plot which is demonstrated in Figure 4. One can conclude that the energy in the Leapfrog method is conserved, therefore, the orbit is stable. This behavior is demonstrated in Figure 5 where the Earth is at a distance $r_0 = 1$ AU and the velocity given by circular motion.

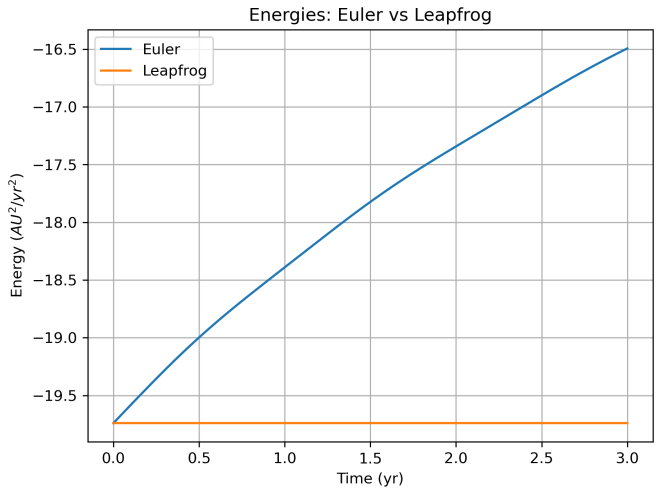
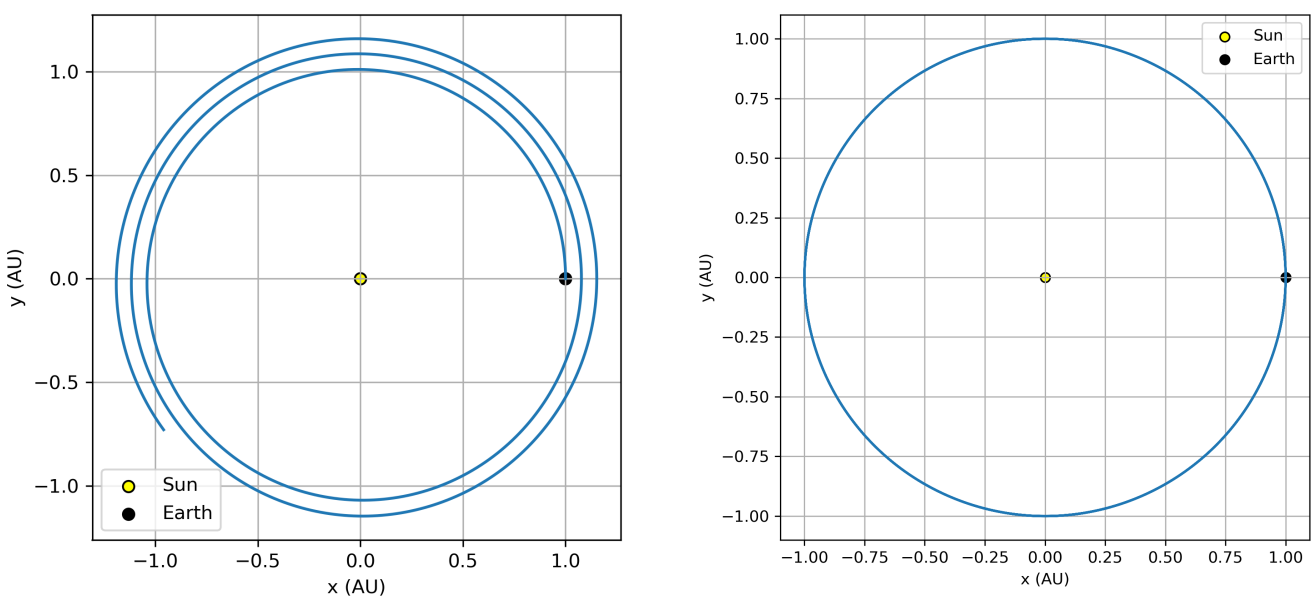


Figure 4. Comparing the energy as a function of time for three orbital periods integrated by the Euler method and by the Leapfrog method.



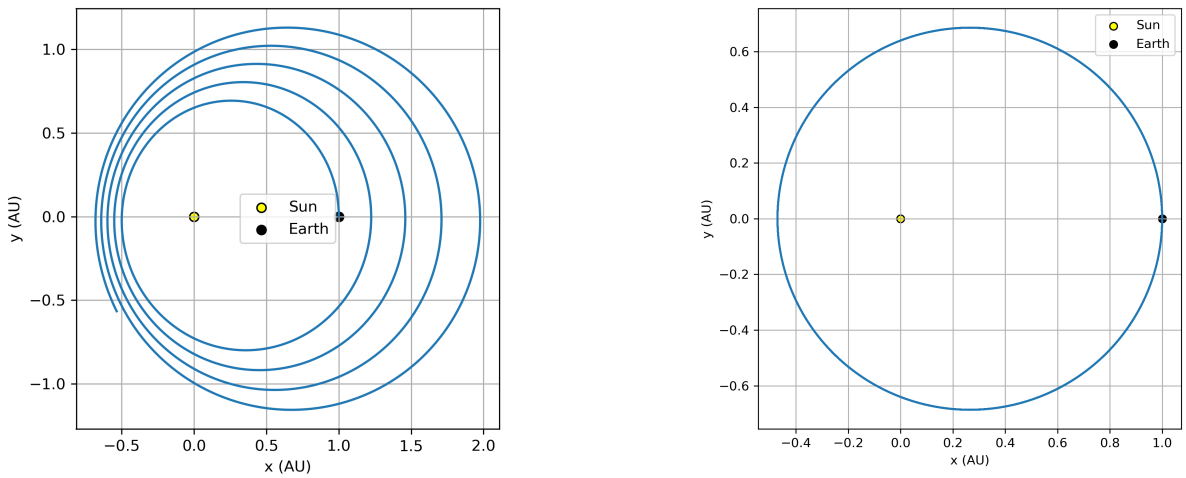
(a) (b)

Figure 5. The orbit of Earth for (a) Euler and (b) Leapfrog integrated over three orbital periods.

2. QUESTION 2

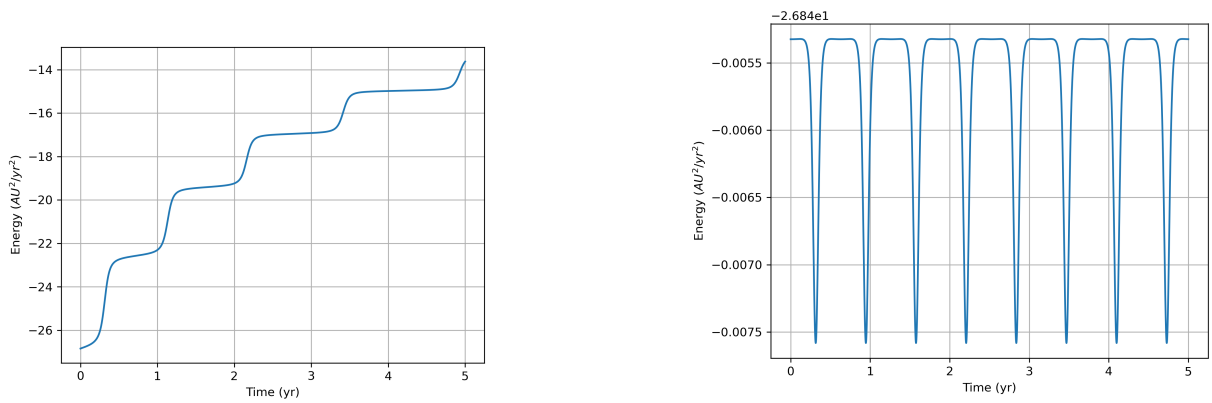
This time, the initial velocity is set to $0.8_{v_{E-orb}}$ for Euler and Leapfrog with results in Figure 6. Again, there is drifting in Euler, but not in Leapfrog. Also, the orbits are not perfectly circular as before. The animations can be found at:

1. Euler: https://github.com/kparrine/Computational_Astro/blob/main/C4/euler5.gif



(a) (b)

Figure 6. The orbit of Earth for (a) Euler and (b) Leapfrog integrated over three orbital periods.



(a) (b)

Figure 7. The energy as a function of time for five orbital periods integrated with the (a) Euler method and with the (b) Leapfrog method.

2. Leapfrog: https://github.com/kparrine/Computational_Astro/tree/main/C4/leapfrog5.gif

In Figure 7 are the total specific energy as a function of time for Euler (a) and Leapfrog (b). For Euler, the energy increases in steps and for Leapfrog it actually decreases sharply before sharply increasing, showing a periodic type effect. Note that this brief change in energy is very small in comparison to Euler. Both have sharp changes in energy because Earth is at perihelion, so the Sun has a larger effect on the speed of Earth at that point such that Earth whips around at that part in its orbit. Another comparison plot is in Figure 8 where Leapfrog's energy looks to be constant.

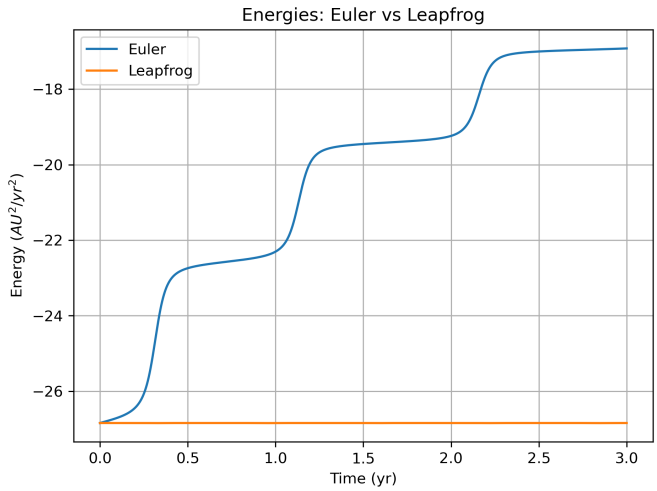


Figure 8. Comparing the energy as a function of time for five orbital periods integrated by the Euler method and by the Leapfrog method.

3. QUESTION 3

To simulate Voyager performing a gravity assist with Jupiter, Jupiter is initialized in a circular orbit with radius $r_J = 5.2$ AU around the Sun and Voyager is launched from Earth ($r_E = 1$) AU. The animation for the Voyager gravitational assist is below:

1. https://github.com/kparrine/Computational_Astro/blob/main/C4/voyager.gif

During the flyby of Voyager, the speed of Voyager as it leaves Earth is 41.37 km/s and as it travels towards Jupiter, the speed of the spacecraft decreases until about 17.5 km/s which is about when the gravitational assist from Jupiter comes into play. After the slingshot, Voyager's velocity sharply increases to 25.91 km/s. For the gravitational assist to happen in this simulation, Voyager was at its closest approach of 0.002 AU at a time of 14.84 months.